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Part One

This part the transformation matrix from the image frame to the room frame is calculated analytically with two approaches.

Approach One

The sequence of transformations that lead to the image frame (I) can be assumed as:

- Translations from O_0 to O_t with $O_0\vec{O}_t = [0.2, 4.1, -0.1]^T$ m
- Rotation corresponding to Euler XYZ angles with $25^\circ, 15^\circ, 3^\circ$ angles respectively
- Translation along X-axis for -0.6 m¹

Therefore, the overall transformation matrix that maps any point in image coordinates to room coordinates is:

$$H_I^0 = T_{x,y,z} R_{x,y,x} T_x \quad (1)$$

where

$$R_{x,y,x}(\alpha, \beta, \gamma) = \begin{bmatrix} R_{x,25^\circ} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_{y,15^\circ} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_{x,3^\circ} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c\beta & s\beta s\gamma & s\beta c\gamma & 0 \\ s\alpha s\beta & -s\alpha c\beta s\gamma + c\alpha c\gamma & -s\alpha c\beta c\gamma - c\alpha s\gamma & 0 \\ -c\alpha s\beta & c\alpha c\beta s\gamma + s\alpha c\gamma & c\alpha c\beta c\gamma - s\alpha s\gamma & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

Here, with $\alpha = 25^\circ$, $\beta = 15^\circ$, and $\gamma = 3^\circ$, the rotation matrix can be calculated as

$$R_{x,y,x}(25^\circ, 15^\circ, 3^\circ) = \begin{bmatrix} 0.9659 & 0.0135 & 0.2584 & 0 \\ 0.1093 & 0.8837 & -0.4550 & 0 \\ -0.2345 & 0.4678 & 0.8521 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

¹This step points out to the transformation of $t \leftarrow I$ denoted as (H_I^t) . Frame I is shifted by -0.6 along the x -axis relative to Frame t ($H_I^t = T_x(-0.6)$). Therefore, to go the opposite way—from Frame I back to Frame t (H_t^I), one must move in the positive x direction by 0.6 meters: $H_t^I = (H_I^t)^{-1} = T_x(+0.6)$.

So,

$$\begin{aligned}
H_I^0 &= \begin{bmatrix} & 0.2 \\ I_{3 \times 3} & 4.1 \\ & -0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix} [R_{x,y,x}]_{4 \times 4} \begin{bmatrix} & -0.6 \\ I_{3 \times 3} & 0 \\ & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 0.9659 & 0.0135 & 0.2585 & -0.3796 \\ 0.1094 & 0.8837 & -0.4551 & 4.0344 \\ -0.2346 & 0.4679 & 0.8521 & 0.0407 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix} \tag{4}
\end{aligned}$$

To transform any point between the two specified frames, we have

$$P_I^I = H_0^I P_0^0 \tag{5}$$

According to equation 5, H_0^I gives the transformation from the image frame to the room frame, which is the objective of this problem. To find H_0^I , we can take the inverse of H_I^0 . For a homogeneous transformation

$$T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \tag{6}$$

the inverse is

$$T^{-1} = \begin{bmatrix} R^T & -R^T p \\ 0 & 1 \end{bmatrix} \tag{7}$$

So, using the above formula, the inverse of H_I^0 can be calculated as

$$H_0^I = (H_I^0)^{-1} = \begin{bmatrix} 0.9659 & 0.1094 & -0.2346 & -0.0651 \\ 0.0135 & 0.8837 & 0.4679 & -3.5791 \\ 0.2585 & -0.4551 & 0.8521 & 1.8994 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix} \tag{8}$$

Approach Two

We need to find the homogeneous transformation matrix H_0^I that maps a point from the operating room frame (Frame 0) to the image/graphics software frame (Frame I) [equation 5]. We can find the transformation by going from Frame 0 to Frame t (the physical target), and then from Frame t to Frame I :

$$H_0^I = H_t^I H_0^t = H_t^I (H_t^0)^{-1} \tag{9}$$

We are given the translation vector and XYX Euler angles for Frame t relative to Frame 0; Thus, the full homogeneous matrix H_t^0 and its inverse using equations 6 and 7 is:

$$H_t^0 = \begin{bmatrix} 0.9659 & 0.0135 & 0.2584 & 0.2 \\ 0.1094 & 0.8837 & -0.4550 & 4.1 \\ -0.2346 & 0.4678 & 0.8521 & -0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{10}$$

$$(H_t^0)^{-1} = \begin{bmatrix} 0.9659 & 0.1094 & -0.2346 & -0.6652 \\ 0.0135 & 0.8837 & 0.4678 & -3.5791 \\ 0.2584 & -0.4550 & 0.8521 & 1.8990 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{11}$$

The corresponding coordinate axes in both Frame t and Frame I are parallel, so there is no rotation ($R_t^I = I_{3 \times 3}$). The origin shifts from one end of the x-axis edge in Frame t to the opposite end in Frame I , so the translation is purely along the x-axis by -0.6 meters:

$$H_t^I = \begin{bmatrix} 1 & 0 & 0 & 0.6 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (12)$$

Finally, by forming equation 9, the same result as in Approach One can be seen.

Part Two

One needs three points to fully determine the position and orientation of the cube. The coordinates of the selected points in the image frame are (figure 1):

$$P_1^I = [0.6, 0, 0.2]^T \text{ m}$$

$$P_2^I = [0, 0.3, 0.2]^T \text{ m}$$

$$P_3^I = [0.6, 0.3, 0]^T \text{ m}$$

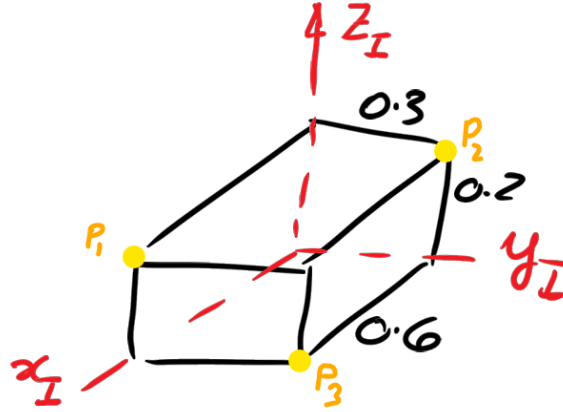


Figure 1: Selected points in the image frame.

However, three individual points do not provide enough spatial information to uniquely solve for all 12 degrees of freedom in a full 4×4 homogeneous transformation matrix if one is going to solve the system of equations in equation 13 without adding the properties of homogenous rotation matrices. Therefore, a fourth point is added to the system by assuming that the origin of the cube in the image frame is located at $P_4^I = [0, 0, 0]^T$.

$$P_i^0 = H_I^0 P_i^I \quad \forall i = 1, 2, 3, 4 \quad (13)$$

We form this equation for the four selected points and solve for H_I^0 in Matlab. Using the analytical forward mapping matrix H_I^0 derived in previous section, one can calculate the position of selected landmarks in Frame 0 ($\vec{P}^0 = H_I^0 \vec{P}^I$). Afterwards, to solve for the 12 unknown parameters of H_I^0 using point correspondences, we isolate each coordinate

dimension (x, y, z) into decoupled linear systems of the form $A X = B$.

$$A = \begin{bmatrix} (\vec{P}_1^I)^T \\ (\vec{P}_2^I)^T \\ (\vec{P}_3^I)^T \\ (\vec{P}_4^I)^T \end{bmatrix} = \begin{bmatrix} 0.6 & 0 & 0.2 & 1 \\ 0 & 0.3 & 0.2 & 1 \\ 0.6 & 0.3 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The target output matrix B collects the measured tracking positions in the room:

$$B = \begin{bmatrix} x_1^0 & y_1^0 & z_1^0 \\ x_2^0 & y_2^0 & z_2^0 \\ x_3^0 & y_3^0 & z_3^0 \\ x_4^0 & y_4^0 & z_4^0 \end{bmatrix} = \begin{bmatrix} 0.4675 & -3.4774 & 2.2249 \\ -0.0792 & -3.2204 & 1.9333 \\ 0.5473 & -3.3059 & 1.9179 \\ -0.0651 & -3.5791 & 1.8994 \end{bmatrix} \quad (14)$$

The answer to the system of equations $A_{4 \times 4} X_{4 \times 3} = B_{4 \times 3}$ is calculated in Matlab with

```
1 H_parameters = (A \ B)';
2 H_I_0 = [H_parameters;
3         0, 0, 0, 1];
```

The code is appended to this report and shows the same result for part two and both approaches of part one.